

HOMEWORK 6 SOLUTIONS

SECTION 3.2.

② a

The determinant is

$$\begin{vmatrix} 1 & 0 & 3 \\ 3 & -4 & c \\ 2 & 5 & 8 \end{vmatrix} \xrightarrow{C_3 - 3C_1} \begin{vmatrix} 1 & 0 & 0 \\ 3 & -4 & c-9 \\ 2 & 5 & 2 \end{vmatrix} = \begin{vmatrix} -4 & c-9 \\ 5 & 2 \end{vmatrix} = -8 - 5(c-9) = -5c + 37$$

Therefore, the matrix is invertible if and only if $-5c + 37 \neq 0$, that is, if $c \neq \frac{37}{5}$

b

The determinant is

$$\begin{vmatrix} 0 & c & -c \\ -1 & 2 & 1 \\ c & -c & c \end{vmatrix} = c^2 \begin{vmatrix} 0 & 1 & -1 \\ -1 & 2 & 1 \\ 1 & -1 & 1 \end{vmatrix} \xrightarrow{R_2 + R_3} c^2 \begin{vmatrix} 0 & 1 & -1 \\ 0 & 1 & 2 \\ 1 & -1 & 1 \end{vmatrix} = c^2 \begin{vmatrix} 1 & -1 \\ 1 & 2 \end{vmatrix} = c^2(2+1) = 3c^2$$

So the matrix is invertible if and only if $3c^2 \neq 0$, or equivalently, if $c \neq 0$

c

The determinant is

$$\begin{vmatrix} c & 1 & 0 \\ 0 & 2 & c \\ -1 & c & 5 \end{vmatrix} \xrightarrow{C_1 - cC_2} \begin{vmatrix} 0 & 1 & 0 \\ -2c & 2 & c \\ -1-c^2 & c & 5 \end{vmatrix} = - \begin{vmatrix} -2c & c \\ -1-c^2 & 5 \end{vmatrix} = c \begin{vmatrix} 2 & 1 \\ 1+c^2 & 5 \end{vmatrix} = c(10 - 1 \cdot c^2) = c(9 - c^2)$$

so the matrix is invertible if and only if $c(9 - c^2) \neq 0$, that is, if $c \neq 0, c \neq -3$ and $c \neq 3$

d

The determinant is

$$\begin{vmatrix} 4 & c & 3 \\ c & 2 & c \\ 5 & c & 4 \end{vmatrix} \xrightarrow{R_3 - R_1} \begin{vmatrix} 4 & c & 3 \\ c & 2 & c \\ 1 & 0 & 1 \end{vmatrix} \xrightarrow{C_3 - C_1} \begin{vmatrix} 4 & c & -1 \\ c & 2 & 0 \\ 1 & 0 & 0 \end{vmatrix} = \begin{vmatrix} c & -1 \\ 2 & 0 \end{vmatrix} = 2 \neq 0$$

so the matrix is invertible for all values of c

③ a

$$\det(A^3 B C^T B^{-1}) = \det(A^3) \det(B) \det(C^T) \det(B^{-1}) = (\det(A))^3 \det(B) \det(C) \cdot \frac{1}{\det(B)} = (-1)^3 \cdot 2 \cdot 3 \cdot \frac{1}{2} = -3$$

b

$$\det(B^2 C^{-1} A B^{-1} C^T) = \det(B^2) \det(C^{-1}) \det(A) \det(B^{-1}) \det(C^T) = (\det(B))^2 \cdot \frac{1}{\det(C)} \det(A) \cdot \frac{1}{\det(B)} \det(C) = 2^2 \cdot \frac{1}{3} \cdot (-1) \cdot \frac{1}{2} \cdot 3 = -2$$

⑤ Let's find $\det(A)$ from the first equation: $-4 = \det(2A^{-1}) = 2^3 \det(A^{-1}) = \frac{8}{\det(A)} \Rightarrow \boxed{\det(A) = -2}$

Now we find $\det(B)$ from the second equation:

$$-4 = \det(A^3(B^{-1})^T) = \det(A^3) \det((B^{-1})^T) = (\det(A))^3 \det(B^{-1}) = \frac{-8}{\det(B)} \Rightarrow \boxed{\det(B) = 2}$$

⑥ a

$$\det(B) = \det(B^T) = \begin{vmatrix} 4u & 4v & 4w \\ 2a & 2b & 2c \\ -p & -q & -r \end{vmatrix} = 4 \cdot 2 \cdot (-1) \cdot \begin{vmatrix} u & v & w \\ a & b & c \\ p & q & r \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_2} = 8 \begin{vmatrix} a & b & c \\ u & v & w \\ p & q & r \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_2} = -8 \begin{vmatrix} a & b & c \\ p & q & r \\ u & v & w \end{vmatrix} = -8 \cdot 3 = -24.$$

Thus, $\det(2B^{-1}) = 2^3 \det(B^{-1}) = \frac{8}{\det(B)} = \frac{8}{-24} = \boxed{-\frac{1}{3}}$

b

$$\det(C) = \det(C^T) = \begin{vmatrix} 2p & 2q & 2r \\ -a+u & -b+v & -c+w \\ 3u & 3v & 3w \end{vmatrix} = 2 \cdot 3 \cdot \begin{vmatrix} p & q & r \\ -a+u & -b+v & -c+w \\ u & v & w \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_3} = 6 \begin{vmatrix} p & q & r \\ -a & -b & -c \\ u & v & w \end{vmatrix} = -6 \begin{vmatrix} p & q & r \\ a & b & c \\ u & v & w \end{vmatrix} \xrightarrow{R_1 \leftrightarrow R_2} = 6 \begin{vmatrix} a & b & c \\ p & q & r \\ u & v & w \end{vmatrix} = 6 \cdot 3 = 18$$

Thus, $\det(2C^{-1}) = 2^3 \det(C^{-1}) = \frac{8}{\det(C)} = \frac{8}{18} = \boxed{\frac{4}{9}}$

SECTION 3.3

① a

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$$

• Characteristic polynomial:

$$C_A(x) = |xI - A| = \begin{vmatrix} x-1 & -2 \\ -3 & x-2 \end{vmatrix} = (x-1)(x-2) - 6 = \boxed{x^2 - 3x - 4}$$

• Eigenvalues: roots of $C_A(x)$:

$$x^2 - 3x - 4 = 0 \Leftrightarrow x = \frac{3 \pm \sqrt{9+16}}{2} = \begin{matrix} 4 \\ -1 \end{matrix}. \text{ The eigenvalues are } \boxed{\lambda_1 = 4, \lambda_2 = -1}$$

• Eigenvector for $\lambda_1 = 4$: nonzero solutions $\vec{x}$ to $(4I - A)\vec{x} = \vec{0}$

$$4I - A = \begin{bmatrix} 3 & -2 \\ -3 & 2 \end{bmatrix} \Rightarrow \text{The eigenvectors are the nontrivial solutions } \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ to } \begin{cases} 3x_1 - 2x_2 = 0 \\ -3x_1 + 2x_2 = 0 \end{cases}$$

Therefore, the 4-eigenvectors are the vectors of the form

$$\boxed{\begin{bmatrix} 3t \\ t \end{bmatrix} \text{ with } t \neq 0}$$

A set of basic 4-eigenvectors is given by $\vec{x}_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

- Eigenvectors for $\lambda_2 = -1$: nonzero solutions to $(-I - A)\vec{x} = \vec{0}$.

$$-I - A = \begin{bmatrix} -2 & -2 \\ -3 & -3 \end{bmatrix} \Rightarrow \text{The } (-1)\text{-eigenvectors are the nontrivial solutions } \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \text{ to } \begin{cases} -2x_1 - 2x_2 = 0 \\ 3x_1 + 3x_2 = 0 \end{cases}$$

So the (-1) -eigenvectors are the vectors of the form

$$\begin{bmatrix} -t \\ t \end{bmatrix}, \text{ with } t \neq 0$$

A set of basic eigenvector is given by $\vec{x}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

Because we have two basic eigenvectors, the matrix A is diagonalisable and $P = \begin{bmatrix} 2 & -1 \\ 3 & 1 \end{bmatrix}$ is a diagonalising matrix. (Invertible because $|P| = 5 \neq 0$)

□

$$A = \begin{bmatrix} 7 & 0 & -4 \\ 0 & 5 & 0 \\ 5 & 0 & -2 \end{bmatrix}$$

- Characteristic polynomial:

$$c_A(x) = \begin{vmatrix} x-7 & 0 & 4 \\ 0 & x-5 & 0 \\ -5 & 0 & x+2 \end{vmatrix} = (x-5) \begin{vmatrix} x-7 & 4 \\ -5 & x+2 \end{vmatrix} = (x-5)(x^2 - 5x - 14 + 20) = (x-5)(x^2 - 5x + 6) = \boxed{(x-5)(x-2)(x-3)}$$

- Eigenvalues: roots of $c_A(x)$. There are $\boxed{\lambda_1 = 2, \lambda_2 = 3, \lambda_3 = 5}$

- Eigenvector for $\lambda_1 = 2$: nontrivial solutions $\vec{x}$ to $(2I - A)\vec{x} = \vec{0}$:

$$(2I - A)\vec{x} = \vec{0} \Leftrightarrow \begin{bmatrix} -5 & 0 & 4 \\ 0 & -3 & 0 \\ -5 & 0 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{4}{5}t \\ 0 \\ t \end{bmatrix}, t \text{ arbitrary}$$

The 2-eigenvectors are $\boxed{\begin{bmatrix} \frac{4}{5}t \\ 0 \\ t \end{bmatrix}}$ with $t \neq 0$. A set of basic eigenvector is given by $\vec{x}_1 = \begin{bmatrix} 4 \\ 0 \\ 5 \end{bmatrix}$

- Eigenvector for $\lambda_2 = 3$: nontrivial solutions $\vec{x}$ to $(3I - A)\vec{x} = \vec{0}$.

$$(3I - A)\vec{x} = \vec{0} \Leftrightarrow \begin{bmatrix} -4 & 0 & 4 \\ 0 & -2 & 0 \\ -5 & 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} t \\ 0 \\ t \end{bmatrix}, t \text{ arbitrary}$$

The 3-eigenvectors are $\boxed{\begin{bmatrix} t \\ 0 \\ t \end{bmatrix}}$, with $t \neq 0$. A set of basic eigenvector is given by $\vec{x}_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

• Eigenvector for $\lambda_3=5$: nontrivial solutions to $(5I-A)\vec{x}=\vec{0}$

$$(5I-A)\vec{x}=\vec{0} \Leftrightarrow \begin{bmatrix} -2 & 0 & 4 \\ 0 & 0 & 0 \\ -5 & 0 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix}, t \text{ arbitrary}$$

The 5-eigenvectors are $\begin{bmatrix} 0 \\ t \\ 0 \end{bmatrix}$, with $t \neq 0$. A set of basic 5-eigenvectors is given by $\vec{x}_3 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$.

Since we have 3 basic eigenvectors, A is diagonalizable and

$$P = \begin{bmatrix} 4 & 1 & 0 \\ 0 & 0 & 1 \\ 5 & 1 & 0 \end{bmatrix} \leftarrow \text{Invertible because } |P| = 1 \neq 0$$

is a diagonalising matrix.

e

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 6 & -6 \\ 1 & 2 & -1 \end{bmatrix}$$

• Characteristic polynomial:

$$c_A(x) = |xI - A| = \begin{vmatrix} x-1 & 2 & -3 \\ -2 & x-6 & 6 \\ -1 & -2 & x+1 \end{vmatrix} \xrightarrow[\substack{C_2 - 2C_1 \\ C_3 + (x+1)C_1}]{\substack{\lambda-1 & 4-2x & x^2-4 \\ -2 & x-2 & -2x+4 \\ -1 & 0 & 0}} = - \begin{vmatrix} 4-2x & x^2-4 \\ x-2 & -2x+4 \end{vmatrix} = -(x-2) \begin{vmatrix} 4-2x & x^2-4 \\ 1 & -2 \end{vmatrix} = -(x-2)^2 \begin{vmatrix} -2 & x+2 \\ 1 & -2 \end{vmatrix} = \boxed{-(x-2)^3}$$

• Eigenvalues: roots of $c_A(x)$. The only eigenvalue is $\boxed{\lambda=2}$. Its multiplicity is 3.

• Eigenvectors of $\lambda=2$: nontrivial solutions $\vec{x}$ to $(2I-A)\vec{x}=\vec{0}$:

$$(2I-A)\vec{x}=\vec{0} \Leftrightarrow \begin{bmatrix} 1 & 2 & -3 \\ -2 & -4 & 6 \\ -1 & -2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Leftrightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3t-2s \\ s \\ t \end{bmatrix}, t, s \text{ arbitrary}$$

The 5-eigenvectors are $\begin{bmatrix} 3t-2s \\ s \\ t \end{bmatrix}$ with $t \neq 0$ or $s \neq 0$. A set of basic eigenvectors is $\vec{x}_1 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}$ and $\vec{x}_2 = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$.

Since the number of basic 5-eigenvectors is smaller than the multiplicity of 5, A is not diagonalizable (i.e. it does not admit a diagonalising matrix P).